

Building footprints from 10 cm aerial imagery

Meta's SAM 3 and Esri's pretrained Mask R-CNN, run across an entire city and scored the same way

Lewisville, Texas • Nearmap imagery, May 2026 • completed September 2026

Python PyTorch + CUDA SAM 3 ArcGIS Pro / ArcPy rasterio / GDAL GeoPandas / Shapely WSL2 GitHub Actions Jupyter

The problem

Cities keep building inventories for permits, taxes and planning, and they go stale: Lewisville's outlines date from 2015 or earlier, while new aerial imagery arrives every year.

The question: how does a general-purpose foundation model, prompted with the word "building", compare with a purpose-built vendor model on the same imagery? And how do you keep that comparison fair and honest when the only reference outlines are years older than the imagery?

What was built

A Python toolkit and CLI that prepares the imagery, runs both models at city scale, applies one shared cleanup, and scores every method against the same reference with the same rules. Settings were tuned on one area under a protocol written down in advance, frozen, and then scored exactly once on a randomly chosen test area.



A typical subdivision: both models outline nearly every house. Aerial imagery: Nearmap, May 2026.

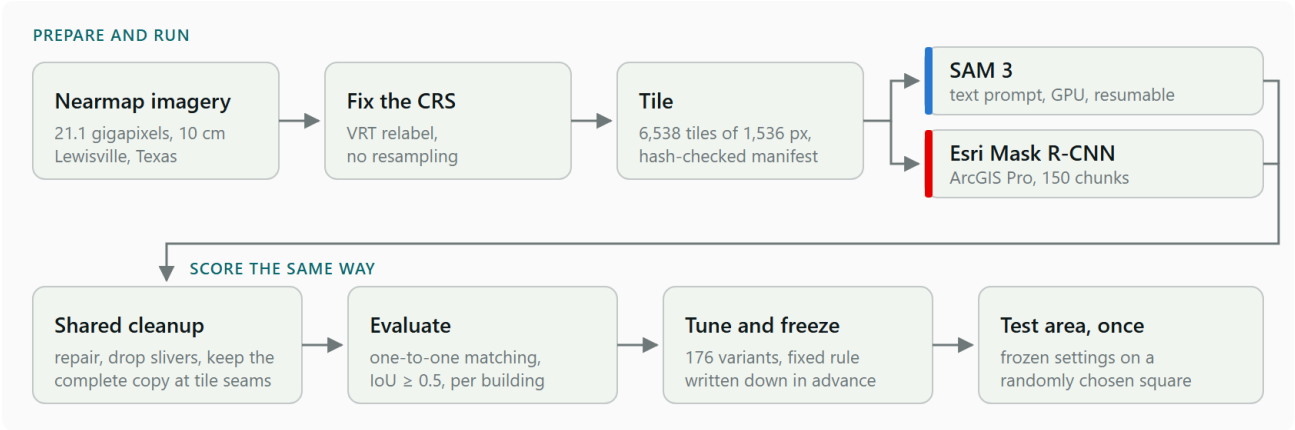
21.1 gigapixels
of 10 cm imagery processed city-wide in 6,538 tiles

0.83 vs 0.74
city-wide F1 on main buildings: Esri vs SAM 3

0.95 · 0.91 · 0.87
test-area F1, scored once: Esri city run, Esri local run, SAM 3

2 h 41 min
SAM 3 over the whole city on a 6 GB laptop GPU

Pipeline



Results

Main buildings of 20 m² and up, against the 2015 outlines	Esri, city run	Esri, local raw run	SAM 3
City-wide F1	0.83	0.77	0.74
City-wide precision / recall	0.82 / 0.85	0.70 / 0.84	0.68 / 0.82
Test-area F1, scored once	0.95	0.91	0.87
Structures under 20 m² found, of 7,867	2	352	267
Run time, whole city	published by the city	8 h 46 min	2 h 41 min

The reference outlines are years older than the imagery, so these scores measure agreement, not accuracy.

What the models do differently



Built since 2015. The 2015 layer has nothing here; both models outline every house.



Carports. Esri outlines the long carport rows that SAM 3 mostly misses.



Large roofs. Long buildings split where SAM 3's tiles meet; Esri's edges step at its chip boundaries.

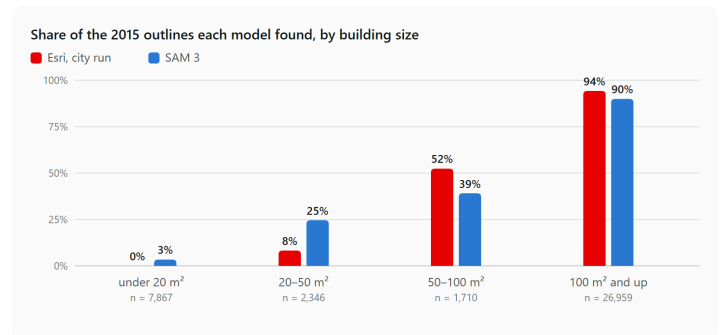
— Esri — SAM 3 — 2015 outlines Aerial imagery: Nearmap, May 2026

Key findings

- **Esri's published run agrees best** with the 2015 outlines, mostly on precision. Its post-processing matters: the same model run locally without it scores 0.77.
- **The models mostly agree:** 91% of Esri's outlines have a matching SAM 3 outline, and SAM 3's matched outlines fit slightly more tightly (median IoU 0.80 against 0.77).
- **Many "extra" outlines are real:** 4,923 SAM 3 outlines with no 2015 counterpart are also drawn by Esri, such as subdivisions built since 2015.
- **Neither finds sheds:** both miss most structures under 20 m².

Engineering highlights

- **CRS repair without resampling:** the 25 GB mosaic was mislabelled as a local metre grid; a VRT relabels it as Texas State Plane in US feet.
- **Resumable, hash-checked inference:** per-tile receipts; a changed tile or output is refused, never silently reused.
- **City-scale Esri runs:** 150 overlapping chunks, each owning a core, so nothing is counted twice. ArcGIS restarts every 8 chunks to release GPU memory that grew until the model failed to load.
- **Fair scoring:** shared cleanup keeps the complete copy of buildings cut at tile seams; one-to-one matching; reports declare whether the reference is a holdout or an older inventory.



Share of the 2015 outlines each model found, by building size.

- **Tuning without peeking:** objective and tie-break rule written in advance, 176 variants, a documented amendment, and a test area the code refuses to score twice.
- **Tested and reproducible:** 151 automated tests in CI on Windows and Ubuntu with Python 3.12 and 3.13; four walkthrough notebooks executed end to end, SAM 3 on the GPU and Esri through ArcGIS Pro.

Next: review outlines for two sample areas against the 2026 imagery to turn agreement into accuracy, with no re-tuning, and fine-tune for sheds.

Links

Repository: github.com/efkopru/nearmap-building-footprints

Full report (PDF, 10 pages): [results/lewisville/Lewisville_building_footprint_comparison.pdf](#)

Results and tuning protocol: [results/lewisville](#) · Notebooks: [notebooks](#)